

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

A veterinarian recommends that each day a certain rabbit should eat **25** calories per pound of the rabbit’s weight, plus an additional **11** calories. Which equation represents this situation, where c is the total number of calories the veterinarian recommends the rabbit should eat each day if the rabbit’s weight is x pounds?

Answer

- A. $c = 25x$
- B. $c = 36x$
- C. $c = 11x + 25$
- D. $c = 25x + 11$

Correct Answer: D

Rationale

Choice D is correct. It’s given that a veterinarian recommends that each day the rabbit should eat **25** calories per pound of the rabbit’s weight, plus an additional **11** calories. If the rabbit’s weight is x pounds, then multiplying **25** calories per pound by the rabbit’s weight, x pounds, yields **25x** calories. Adding the additional **11** calories that the rabbit should eat each day yields **25x + 11** calories. It’s given that c is the total number of calories the veterinarian recommends the rabbit should eat each day if the rabbit’s weight is x pounds. Therefore, this situation can be represented by the equation $c = 25x + 11$.

Choice A is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat **25** calories per pound of the rabbit’s weight.

Choice B is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat **25 + 11**, or **36**, calories per pound of the rabbit’s weight.

Choice C is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat **11** calories per pound of the rabbit’s weight, plus an additional **25** calories.